

Rewards360 — Master Generation Prompt (React UI, Module 2.1)

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Purpose: Save and reuse this prompt to regenerate the full **Rewards360** React UI for **Module 2.1 – Customer Registration & Profile Management**, aligned to Section 4.1. It covers Registration, Login (Password & OTP), Forgot Password, Customer Profile Dashboard, Preferences, **Profile Picture upload**, accessibility, and a cohesive **light-green** theme.

Tech & Architecture Constraints

- **React 18** with **JavaScript + JSX** (no TypeScript).
- **React Router v6** with routes: `/register`, `/login`, `/dashboard`.
- **CSS Modules** for all styling; **no external UI libraries**.
- Functional components + hooks (`useState`, `useEffect`).
- Accessibility-first: semantic HTML, labels, `aria-*`, focus management, `aria-live` banners.

Theme & Layout Requirements

- Use a single **light-green** palette via CSS variables:

CSS ▾

```
1 :root {
2   --brand: #9FE2BF;           /* light green */
3   --brand-dark: #2E8B57;      /* darker green for header/buttons */
4   --brand-contrast: #ffffff;   /* text on brand areas */
5   --brand-gradient-start: #AEEFCB;
6   --brand-gradient-end: #86D9AD;
7 }
8
```

- **Header:** green bar with “**Rewards360**” on the left and nav links **Register | Login |**

Profile on the right.

- **Auth pages** (Register & Login): centered two-panel card
 - **Left:** "Welcome" panel with green gradient & outline button.
 - **Right:** page title + form card.
 - **Points Summary tiles:** label on top, value below; centered; label `(30d)` must remain on one line (`white-space: nowrap`) and tiles should have adequate min-width or use fluid grid.
 - **Purchase History** table: center-aligned cells, readable header.
 - Ensure **consistent vertical spacing** (margin) above Purchase History equal to spacing between Points Summary and Preferences.
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Files to Generate (with complete code blocks)

CSS ▾

```
1  src/
2    App.jsx
3    pages/
4      RegisterPage.jsx
5      LoginPage.jsx
6      ProfileDashboard.jsx
7    components/
8      RegistrationForm.jsx
9      LoginForm.jsx
10     ForgotPassword.jsx
11     OtpInput.jsx
12     ProfileHeader.jsx
13     PointsSummary.jsx
14     PurchaseHistory.jsx
15     PreferencesPanel.jsx
16     EditProfileModal.jsx
17     ChangePasswordModal.jsx
18   services/
19     api.js
20   validators/
21     index.js
22   styles/
23     App.module.css
24     Form.module.css
25     OtpInput.module.css
26     Panel.module.css
27     AuthCard.module.css
28     Profile.module.css
29   README.md
30
```

Output format: JavaScript + JSX only (no TS). Include a short **README** (run steps +

App Shell & Routing

- `src/App.jsx`: Header (brand left, nav right), global `aria-live` banner, router `<Routes>` for `/register`, `/login`, `/dashboard`, and a footer ("Reward 360 — Customer Registration & Profile Management").
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Screen Specifications

1) Register (Customer Registration)

- **Fields:** Name, Email, Phone, Password; Preferences:
 - Favorite categories (multi-select): Fashion, Electronics, Groceries, Beauty, Home.
 - Communication mode (radio): Email, SMS, WhatsApp.
 - **Profile Picture (optional):** file input with preview; store Data URL in demo persistence.
- **Validation:**
 - Name ≥ 2 chars; Email (regex); Phone (10–15 digits); Password ≥ 8 chars with upper+lower+digit+special (`!@#$$%^&*`).
- **UX:** Inline errors; disable submit until valid; show/hide password; proper tab order; `aria-invalid` + `aria-describedby`; focus first invalid field on submit.
- **Submit:** Call `POST /api/auth/register` (stubbed in `services/api.js`) with payload `{ name, email, phone, password, preferences: { favoriteCategories, communicationMode }, photoURL }`. Success $\rightarrow$ banner + redirect `/login`; failure $\rightarrow$ error banner.

2) Login (Secure)

- **Modes:** Password login (Email + Password) and Login with OTP (Email $\rightarrow$ request OTP $\rightarrow$ verify $\rightarrow$ token).
- **Forgot Password** panel:
 - Step 1: Email $\rightarrow$ `POST /api/auth/request-otp`
 - Step 2: 6-digit OTP via **OtpInput** (arrow navigation, backspace behavior, paste full code).

- Step 3: Reset Password → `POST /api/auth/reset-password` (same strength rules).
- **UX:** Loading states; accessible alerts; `aria-live="polite"` for messages; demo token saved in `sessionStorage` with comment to prefer HTTP-only cookies in production.

3) Customer Profile Dashboard

- **Header:** avatar (use uploaded `photoURL`, fallback initials), name, email, phone, loyalty tier badge.
 - **Points Summary:** tiles for Balance, Earned (30d), Redeemed (30d), Expiring soon — label above value; labels must not wrap; tiles centered with fluid grid/min-width.
 - **Saved Preferences:** view & update favorite categories + communication mode.
 - **Purchase History:** accessible table (Date, Order ID, Amount, Points Earned, Points Redeemed) with **center-aligned** cells.
 - **Actions:**
 - **Edit Profile** modal (name, phone, **profile picture** upload/change/remove; email treated as key).
 - **Change Password** modal (current/new, show/hide, validation).
 - **Spacing:** Add margin above Purchase History to match spacing between Points and Preferences.
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Component Requirements

- `RegistrationForm.jsx`: controlled inputs, validators, avatar preview, payload includes `photoURL`.
- `LoginForm.jsx`: Password/OTP mode switch styled in green; validation; banner messaging; navigate to dashboard.
- `ForgotPassword.jsx`: 3-step flow; `OtpInput` ref exposes `focusFirst()`; accessible status text.
- `OtpInput.jsx`: 6 fields; arrow left/right; backspace; full paste; `inputMode="numeric"`.
- `ProfileHeader.jsx`: image avatar if `photoURL`; otherwise initials.
- `PointsSummary.jsx`: stacked tiles, `white-space: nowrap` on labels; fluid grid/min-width for `(30d)`.
- `PurchaseHistory.jsx`: center aligned `th`/`td`.
- `PreferencesPanel.jsx`: updates preferences via callback.
- `EditProfileModal.jsx`: change/remove photo (Data URL), persist on save.

- `ChangePasswordModal.jsx`: show/hide; validation; success banner.
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Accessibility & UX Rules

- Every input has a `<label>` (`htmlFor`) and matching `id`.
 - Errors use `aria-describedby` and `role="alert"`.
 - Global banner uses `aria-live="polite"`.
 - Focus first invalid field on submit; focus first OTP field when panel opens.
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Validators (module)

- `isEmail(value)` → basic email regex.
 - `isPhone(value)` → `/^\d{10,15}$/`.
 - `isPasswordStrong(value)` → must include upper, lower, digit, special; length ≥ 8 .
 - `isOtp(value)` → `/^\d{6}$/`.
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API Layer (Stubbed) & Demo Persistence

- `src/services/api.js` uses `localStorage` for demo persistence:
 - `register(payload)` → saves `{name,email,phone,preferences,photoURL,loyaltyTier:'Bronze'}` keyed by email.
 - `login({email,password})` / `verifyOtp({email,otp})` → set `r360_current_user`, return demo token.
 - `requestOtp({email})`, `resetPassword({email,otp,newPassword})` → simulated success.
 - `getProfile()` → returns current user's profile (incl. **photoURL**).
 - `getPointsSummary()`, `getPurchaseHistory()` → demo data.
 - `updateProfile({ name, email, phone, photoURL })`, `updatePreferences(preferences)`, `changePassword(...)` → update demo store.

Production Notes: Replace stubs with real endpoints, use **HTTPS**, **HTTP-only secure cookies**, server-side validation, **CORS/CSRF** protections.

Acceptance Criteria

- Registration validates and redirects to Login.
 - Login supports Password & OTP; Forgot Password runs Email → OTP → Reset.
 - Dashboard shows profile (incl. photo), loyalty tier, points summary, purchase history, preferences; supports Edit Profile & Change Password.
 - OTP input supports arrows, backspace, paste.
 - All inputs accessible; forms responsive; cohesive **light-green** theme.
 - Code is modular, commented, backend-ready.
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Styling (CSS Modules) – Key Rules

- Global `box-sizing: border-box` to prevent overflow.
 - Header `.header` uses `var(--brand-dark)`; `.nav` aligned right via `margin-left: auto`; `.brand` shows “Rewards360”.
 - Auth card: left gradient panel + right pale green panel; centered via `.panelCenter`.
 - Forms: `.actions { justify-content: flex-end; }`.
 - Points tiles: `.pointsLabel { white-space: nowrap; }`; tiles with min-width or fluid grid: `repeat(auto-fit, minmax(220px, 1fr))`.
 - Table cells: `th, td { text-align: center; }`.
 - Add vertical margin before Purchase History (e.g., `section[aria-label='Purchase history'] { margin-top: 1rem; }`).
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README Snippet to Include

- Run locally: `npx create-react-app reward360-ui`, replace `src/`, `npm install react-router-dom@6`, `npm start`.
 - Backend wiring: example `fetch` / `axios` snippets (cookie auth), and security notes.
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Final Generation Prompt (Paste into Copilot)

 Prompt:

Generate a complete **Rewards360** React 18 UI (JavaScript + JSX, no TypeScript) for **Module 2.1 – Customer Registration & Profile Management** per Section 4.1. Use **React Router v6** and **CSS Modules**, no external UI libraries. Build the exact files listed (App.jsx, pages, components, services/api.js, validators, styles, README). Apply the **light-green** theme via CSS variables (brand colors above) and ensure accessibility.

Screens:

1. **Register:** Name, Email, Phone, Password; Preferences (multi-select categories; radio communication); **Profile Picture (optional)** preview. Validate per rules; inline errors; show/hide password; focus first invalid field. Submit to stubbed `POST /api/auth/register`. Success banner → `/login`.
2. **Login:** Password login & **Login with OTP**. Forgot Password flow: Email → `POST /api/auth/request-otp`; 6-digit `OtpInput` (arrows, backspace, paste) → verify → Reset Password (`POST /api/auth/reset-password`). Use `aria-live` banners; save demo token in `sessionStorage` with comments recommending **HTTP-only cookies** in prod.
3. **Profile Dashboard:** header with avatar (use `photoURL` if set, initials fallback), name, email, phone, loyalty tier. **Points Summary** tiles (Balance, Earned (30d), Redeemed (30d), Expiring soon) with **label above value**; labels must **not wrap**; tiles centered with fluid grid/min-width. **Saved Preferences** editor. **Purchase History** table with center-aligned cells. Actions: **Edit Profile** (incl. photo upload/change/remove) & **Change Password** modal.

API layer: `src/services/api.js` with demo persistence (`localStorage`): `register`, `login`, `requestOtp`, `verifyOtp`, `resetPassword`, `getProfile`, `getPointsSummary`, `getPurchaseHistory`, `updateProfile`, `updatePreferences`, `changePassword` (store `photoURL`, set `r360_current_user`). Comment where real backend replaces stubs (HTTPS, **HTTP-only cookies**, CORS/CSRF, server validation).

Validators: `isEmail`, `isPhone`, `isPasswordStrong`, `isOtp` in `src/validators/index.js`.

Styles: `App.module.css`, `Form.module.css`, `OtpInput.module.css`, `Panel.module.css`, `AuthCard.module.css`, `Profile.module.css`; include global `box-sizing`, centered auth cards (`.panelCenter`), form actions right-aligned, table cells centered, margin before Purchase History, points tiles stacked & fluid.

Acceptance Criteria: Registration validates/redirects; Login password & OTP work; Forgot Password email→OTP→reset works; Dashboard shows profile/tier/points/history/preferences; Edit Profile & Change Password work; OTP input interactions work; accessible forms; cohesive light-green theme; modular code ready for backend.

Optional Follow-up Prompts (Backend Wiring)

- **Upload backend wiring:** Replace Data URL avatar with real upload (Spring Boot / ASP.NET Core). Provide endpoints for photo upload, storage, and URL return; paste-ready frontend `fetch` / `axios` (with progress), and security notes (S3/Blob Storage, CORS, type/size validation).
- **Production auth:** Switch to cookie-based auth; show login/register/reset with HTTP-only cookies, CSRF protection, and secure CORS.